

QUESTION 1.

2

Marks

a) Factorise fully i) $9x^2 - 49$

ii) $m^3 - 9m^2 + 3m - 27$

2

b) Solve $3n - 7 \leq 5 - n$

2

c) Differentiate with respect to t : $y = (5t^3 - 3)^7$

2

d) Evaluate exactly: $\sin 135^\circ + \tan 120^\circ$

2

e) Find a primitive for $3x^2 + \cos 5x$

2

f) Evaluate $\log_5 8$ correct to 4 significant figures

1

g) Evaluate: $|-7 - 8| - |3 - 11|$

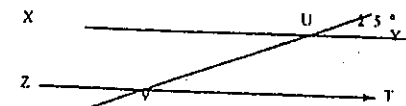
1

QUESTION 2 (Start a new page)

3

Marks

a)



2

In the diagram above, $XY \parallel ZT$.Find the size of $\angle UVZ$ and give a reason for your answer.

b) On a number plane, sketch the region which is described by

$x^2 + y^2 \leq 9$ and $y > x^2 + 1$

3

c) Differentiate the following functions with respect to x :

i. $7 \tan x$;

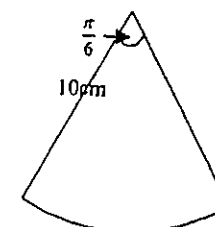
5

ii. $\frac{x-1}{3x-4}$;

iii. $x e^{\sin x}$

d)

2



NOT TO SCALE

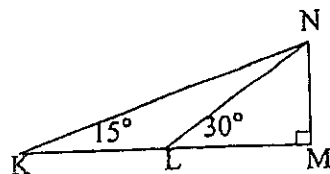
A sector drawn in a circle of radius 10cm has a sector angle of $\frac{\pi}{6}$ radians. Find the length of the arc correct to 3 decimal places.

QUESTION 3 (Start a new Page)

Marks

- a) Fifty identical cards numbered from 1 to 50 are placed in a bag, and one card is drawn at random. What is the probability that this card will be either less than 20 or divisible by 3? 2

b)



- Find $\angle LNM$, $\angle KNL$ in degrees, and show that $KL = LN$. 2
- Given that the length of NM is 1 unit, find the *exact* lengths of LM and LN . 2
- Deduce that $\tan 15^\circ = 2 - \sqrt{3}$. 1

- b) Find the following indefinite integrals: 5

- $\int 3 \cos 2x dx$;
- $\int \frac{4x+5}{2x^2+5x} dx$;
- $\int \frac{x+1}{\sqrt{x}} dx$.

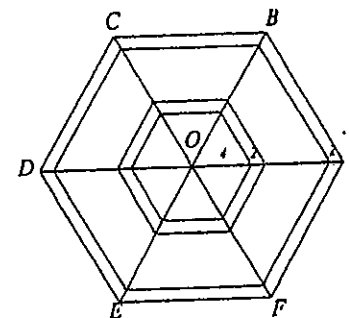
QUESTION 4 (Start a new page)

Marks

- (a) Write down the formula for:
 (i) the n th term of an arithmetic series with first term a and common difference d ; 1
 (ii) the sum of the first n terms of this series. 1

A particular spider's web consists of a series of regular hexagons with a common centre O , held together by rays through O , as in the figure, where only some of the hexagons are shown.

NOT TO SCALE, LENGTHS IN CM.



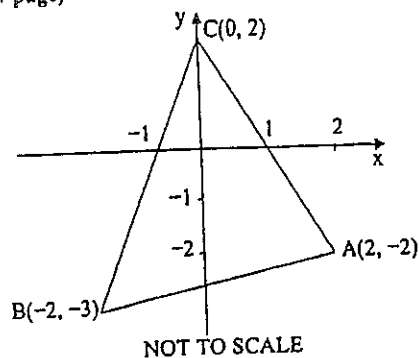
The vertices of the smallest hexagon are 4 cm from O , the vertices of the next are 2 cm further away and they continue at 2 cm intervals along the rays until the vertices of the last hexagon ABCDEF are 60 cm from O (i.e. $OA = 60$ cm).

- Using parts (i) and/or (ii), show that there are 29 hexagons in the spider's web. 1
 - What is the length, in cm, of the perimeter of the smallest hexagon? 2
 - What is the total length of thread used by the spider in making this web (including the six rays from O)? 3
- b) A function $y = f(x)$ has a stationary point at $(2, -6)$ and $f''(x) = 6x - 4$
- Find
- the nature of the stationary point at $(2, -6)$ 1
 - the equation of the function 3

136

QUESTION 5 (Start a new page)

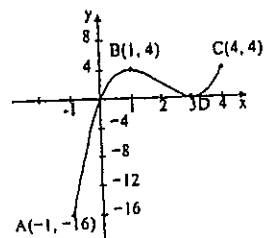
Marks



A(2, -2), B(-2, -3) and C(0, 2) are the vertices of a triangle ABC.

- Find the length of the interval AC
- Find the gradient of AC.
- Show that the equation of the line AC is $y = -2x + 2$.
- Calculate the perpendicular distance of B from the side AC
- Hence find the area of $\triangle ABC$.
- Find the co-ordinates of D such that ABCD is a parallelogram

b)



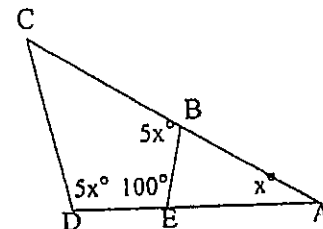
- Using the words positive, negative or zero, complete the following sentences:
At B and D, $f'(x)$ is _____. (answer on your answer paper).
Between D and C, $f'(x)$ is _____. (answer on your answer paper).
- On Page 12, the above diagram is reproduced. On the axes underneath it, sketch the graph of $y = f'(x)$

QUESTION 6 (Start a new page)

Marks

a) Evaluate $\lim_{x \rightarrow 3} \frac{x^3 - 27}{x - 3}$.

b)

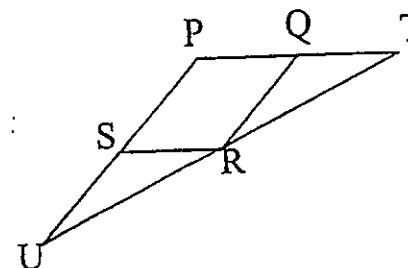


As shown in the figure (which is not to scale), B and E lie on the sides AC and DA respectively of $\triangle ACD$.

Use the information shown on the figure to

- find the value of x
- and hence give $\angle ACD$ in degrees.
Give reasons for your answers.

c)



PQRS is a parallelogram. PQ is produced beyond Q to T so that $QT = QR$ and PS is produced beyond S to U so that $SU = PS$. T, R and U are collinear.
Prove that PQRS is a rhombus.

- If α and β are the roots of $2x^2 + 3x - 4 = 0$, find the value of:
 - $\alpha + \beta$;
 - $\alpha^2 + \beta^2$.
- Solve $2x^2 + 3x - 2 < 0$

QUESTION 7 (Start a new page)

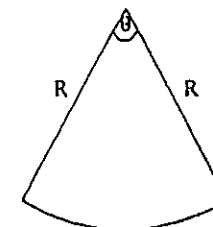
Marks

- a) A parabola has equation $x^2 + 6x - 33 = 12y$. Find:
- the axis of symmetry
 - the coordinates of the vertex;
 - the coordinates of the focus;
 - the equation of its directrix.
- 4
- b) A ship sails from port A on a course of 075° for 20 nautical miles then changes its course to 130° and continues sailing for 30 nautical miles.
- Draw a neat sketch of the ship's course. 1
 - How far is it from its starting point? (Answer to the nearest nautical mile). 1
 - What is the ship's bearing from its starting point? (Answer to the nearest degree). 2
- c) Use
- the Trapezoidal rule and
 - Simpson's Rule to estimate $\int_1^4 \log_e(x^2) dx$, (one application of each) rounding your answer to one decimal place. 4

QUESTION 8 (Start a new page)

Marks

- a) For $\frac{\pi}{2} < A < \frac{3\pi}{2}$, find A if $\sin^2 A = \frac{1}{4}$. 2
- b) Two ordinary dice are tossed. What is the probability that
- the uppermost faces will show a pair of "5's" 3
 - the uppermost faces will show exactly one "3" and one "5"
 - the uppermost faces of at least one die will show a "5"
- c)
- d)



NOT TO SCALE

If the area of this sector is 625 m^2 ,

- show that $\theta = \frac{1250}{R^2}$, and 1
 - find an expression for P (the perimeter) in terms of R and θ , and hence show that 2
- $$P = 2R + \frac{1250}{R}$$

- Find the values of R and θ (to the nearest degree) so that the sector has minimum perimeter. 2

QUESTION 9 (Start a new page)

Marks

- a) Evaluate $\sum_{k=2}^8 2^k$ 2
- b) The rate at which water runs out of a tank is proportional to the volume of water in the tank, i.e. $\frac{dV}{dt} = kV$. The tank is full to start with and has a capacity of 36 000 litres.
- i) Show that $V = V_0 e^{kt}$ satisfies this equation where V_0 is the volume of water in the tank initially. 1
- ii) If $\frac{1}{4}$ of the water in the tank runs out in 30 minutes, find the volume of water remaining in the tank after 60 minutes. 3
- c) The area under the curve $y = \sqrt{9 - x^2}$, $-3 \leq x \leq 3$, is rotated about the x axis.
- i) Find the volume of the solid of revolution thus obtained. 3
- ii) Describe the shape of this solid. 1
- d) The sum of the first n terms of a series is given by $S_n = 3^n + 2n^2$. Find the 13th term. 2

QUESTION 10 (Start a new page)

Marks

- a) Solve for x: $2\ln x - \ln(2 - x) = \ln 2 - \ln 3$ 3
- b) i) Expand $e^{-x}(1 - e^{-x})$. 1
- ii) For the curve $y = e^{-x} - e^{-2x}$
- a) Show that it cuts the axes at (0, 0) only. 1
- b) Show that there is only one stationary point at $(\ln 2, \frac{1}{4})$, and determine the nature of that stationary point. 3
- c) Hence determine the values of x for which the curve has a negative gradient, and so discuss the behaviour of the curve for large values of x. 1
- d) Sketch the curve. 1
- e) Calculate the area bounded by this curve, the x-axis and the ordinates $x = 0$ and $x = \ln 2$. 2

End of Paper

Q1

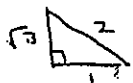
a) (i) $(3x-7)(3x+7)$ (1)

(ii) $m^2(m-9) + 3(m-9)$
 $= (m-9)(m^2+3)$ (1)

b) $3n-7 \leq 5-n$ (1)
 $4n \leq 12$ (1)
 $n \leq 3$ (1)

c) $y = (5t^3-3)^7$ (1)
 $y' = 7(5t^3-3)^6 \times 15t^2$ (1)
 $= 105t^2(5t^3-3)^6$ (1)

d) $\tan 135^\circ + \tan 120^\circ$
 $= \tan 45^\circ - \tan 60^\circ$ (1)
 $= \frac{1}{\sqrt{2}} - \sqrt{3}$ (1)



e) $\int (3x^2 + \cos 5x) dx = x^3 + \frac{1}{5} \sin 5x + c$ (1)

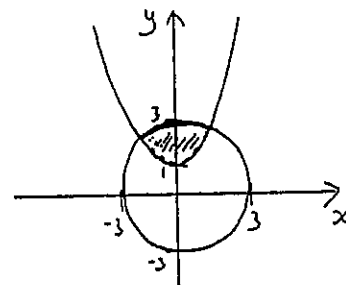
f) $\log_5 8 = \frac{\log 8}{\log 5} = 1.292$ (1)

g) $|-7-8| - |3-11| = 15-8 = 7$ (1)

Marking Scheme Q1

a) ① $\angle UVZ = 155^\circ$

① pair of reasons (e.g. vertically opp. angles equal
 co-interior angles $XY \parallel ZT$)



① circle, centre origin, radius 3

① parabola, concave up, vertex at $y=1$

② correct region is indicated

② circle edge included,
 parabola edge is dashed

c) i) $y' = 7 \sec^2 x$

$\sec^2 x$ starts (1/2)

numerator x (using incorrect denominator $\sqrt{1/2}$)

ii) $y' = \frac{-1}{(3x-4)^2}$

numerator $\sqrt{}$, denom. x (1)

numerator x (using correct $\sqrt{}$ but denom $\sqrt{}$) (1)

numerator x (using incorrect $\sqrt{}$ but denom $\sqrt{}$) (1/2)

iii) $y' = \frac{e^{\sin x} + x \cos x e^{\sin x}}{e^{\sin x} (1 + x \cos x)}$

$y' = x \cos x e^{\sin x}$ (1/2)

no product rule (1)

product rule and $\cos x$ appears somewhere and $e^{\sin x}$ appears somewhere (1)

product rule and no $\cos x$ and no $e^{\sin x}$ mistake (1)

d) $L = 10 \times \frac{\pi}{6} = 5.23598$
 $L = 5.236 \text{ cm}$ (3 d.pl.)
 rounded correctly to 3 d.pl.

5.235 s

140

② a)

Factorising the top correctly
 Cancelling (x-3) top and bottom
 Substituting: x=3, to get 27

minors are for each bit missing

(1.5) only 2

③ b)

QTR is isosceles → 1/2
 or
 180 + half each
 180 - 5x = 180
 angles ... straight line
 angle sum of triangle
 x = 20

① $\hat{BCD} + 5x + x = 180^\circ$ (angle sum of $\Delta = 180^\circ$)
 ① $\hat{BCD} + 5x + 5x + 100 = 360^\circ$ (angle sum of quadrilateral = 360)
 ① Solve the above simultaneously (or attempting)
 ① Finding $x = 20^\circ$
 ① [Substituting $x = 20^\circ$ to obtain $\hat{BCD} = 60^\circ$

③ c)

① Explaining that QRT is an isosceles triangle
 ① Explaining that SUR is also isosceles
 ① Explaining that adjacent sides of PQRS are therefore the same

② d)

① $\alpha + \beta = -3/2$
 ① $\alpha^2 + \beta^2 = \frac{25}{4}$ or $6\frac{1}{4}$

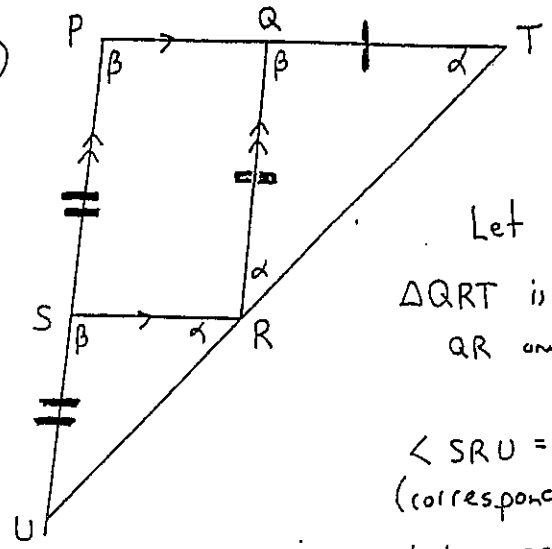
Correct formula and substitution but incorrect answer given ②

② e)

minors one for each bit missing

Factorising correctly $(2x-1)(x+2)$
 Finding x values correctly ($\frac{1}{2}$ and -2)
 $-2 < x < \frac{1}{2}$
 $x > -2$ and $x < \frac{1}{2}$ says ②
 $x < -2$ and $x < \frac{1}{2}$ says ①

c)



Let $\angle QTR = \alpha$
 ΔQRT is isosceles (2 sides equal, QR and QT)

$\angle SRU = \alpha$
 (corresponding angles, $PT \parallel SR$)

Let $\angle QPS = \beta$
 $\therefore \angle PQR$ and $\angle RSU$ are both also equal to β
 (corresponding angles, parallel lines)

$\therefore \Delta SRU$ is similar to ΔQTR (AA)
 $\therefore \Delta SRU$ is isosceles
 $\therefore SU = SR$ (sides of isosceles triangle)

But $PS = SU$ (given)

$\therefore PS = SR$

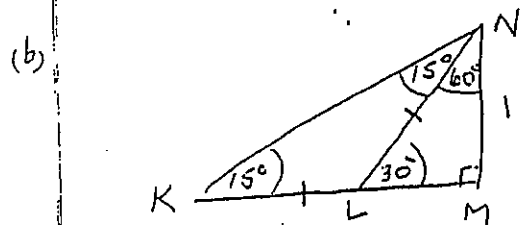
$\therefore PQRS$ must be a rhombus (parallelogram with adjacent sides equal)

Q.E.D. !

Q3

$$P(x < 20 \text{ or } x \geq 3) = \frac{19 + 16}{50} = \frac{35}{50} = \frac{7}{10} \quad (2)$$

$\frac{35}{50} \rightarrow 1$



(i) $\frac{\angle LNM}{\angle KNL} = \frac{60^\circ}{180 - (90 + 60 + 15)} = 15^\circ$

Since $\angle LKN = \angle KNL$, $KL = LN$
(base angles of isosceles $\triangle KLN$)

(ii) $\sin 30^\circ = \frac{1}{2} = \frac{NM}{NL} = \frac{1}{NL}$

$\therefore \frac{NL}{\sin 60^\circ} = \frac{KL}{\sin 30^\circ} = \frac{LM}{\sin 15^\circ}$

(iii) $\tan K = \frac{NM}{KM} = \frac{1}{2 + \sqrt{3}}$

$= \frac{1}{2 + \sqrt{3}} \times \frac{2 - \sqrt{3}}{2 - \sqrt{3}} = \frac{2 - \sqrt{3}}{4 - 3} = 2 - \sqrt{3}$

(b) (i) $\int 3 \cos 2x \, dx = \frac{3}{2} \sin 2x + C \quad (2)$

(ii) $\int \frac{4x+5}{2x^2+5x} \, dx = \ln(2x^2+5x) + C \quad (1)$

(iii) $\int \frac{2x+1}{\sqrt{3x}} \, dx = \int (x^{\frac{1}{2}} + x^{-\frac{1}{2}}) \, dx$

$= \frac{2}{3} x^{\frac{3}{2}} + 2x^{\frac{1}{2}} + C \quad (2)$

Question (4)

(i) $T_n = a + (n-1)d \quad (1)$

(ii) $S_n = \frac{n}{2} (2a + (n-1)d) \quad (1)$

(iii) $a = 4, d = 2, T_n = 60$

$60 = 4 + (n-1) \cdot 2$

$56 = 2n - 2$

$2n = 58$

$n = 29 \quad (1)$

(iv) $x = 4$

$4 \times 6 = 24 \text{ cm} \quad (2)$

(v) hex's = $24 + 36 + \dots + 360$

Rays = $6 \times 60 = 360$

total = $\frac{29}{2} [24 + 360] + 360$

$= 5928 \text{ cm} \quad (3)$

(i) $f''(x) = 6x - 4$

(ii) $f''(2) = 8 > 0 \quad (1)$

$\therefore$ minimum Turning Pt

(iii) $f'(x) = 3x^2 - 4x + C$

$f'(2) = 0$

$0 = 12 - 8 + C$

$C = -4$

$\therefore f'(x) = 3x^2 - 4x - 4$

$f(x) = x^3 - 2x^2 - 4x + K$

$f(2) = -6$

$-6 = 8 - 8 - 8 + K$

$K = 2$

$\therefore f(x) = x^3 - 2x^2 - 4x + 2$

Q5 a)

i) $AC = \sqrt{20}$ or $2\sqrt{5}$

out of 12

ii) $m_{AC} = -2$

iii) substituting into $y = mx + b$ or $y - y_1 = m(x - x_1)$

$A(2, -2)$ or $C(0, 2)$

iv) $d = \frac{|2 \times -2 + 3 \times -1 + -2|}{\sqrt{2^2 + 1^2}}$

$= \frac{9}{\sqrt{5}}$ or $\frac{9\sqrt{5}}{5}$

v) $A = 9 \cdot 0^2$ or $\frac{1}{2} \times (i) \times (iv)$

vi) D is at $(4, 3)$

b)

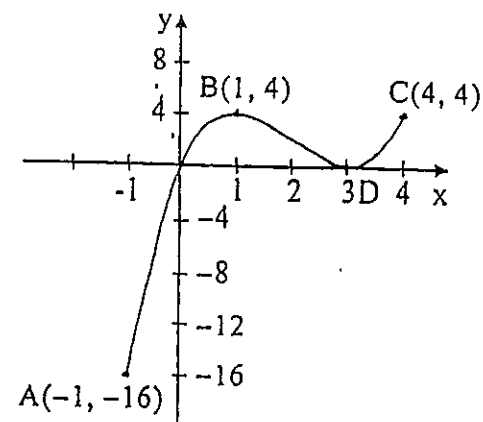
i) ~~positive~~ zero

~~positive~~

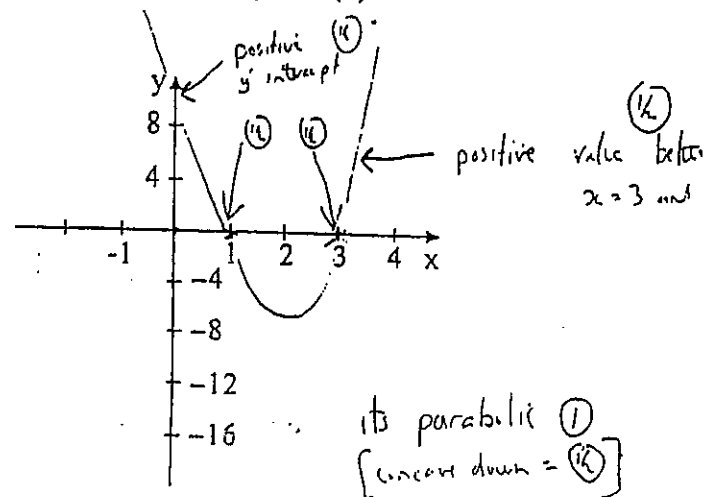
ii) on sheet

Question 5 (b)

Graph of $y = f(x)$



Graph of $y = f'(x)$



$$y = \frac{1}{12}x^2 + \frac{1}{2}x - 2\frac{3}{4}$$

(1)

(7)

20

(1)

①

2.

1)

(2)

14.

(2)

(144)

a)

$\frac{1}{2}$ for $\pm$ if they don't score
for different answers below

(minus half if answer
in degrees 150, 210)

i)

ii)

iii)

$2/36$ is okay

ignore

i)

ii).

[substitution of expression for θ]

iii)

[differentiating only]
scores (Y₂)

①

①

115 degrees

is
fine

(a) $\sum_{k=2}^8 2^k = 2^2 + 2^3 + 2^4 + \dots + 2^8$
 $S_7 = \frac{4(2^7 - 1)}{2 - 1} = 4 \times (128 - 1)$ ①
 $= 508$ ①

(b) $\frac{dV}{dt} = kV$
 $\therefore V = V_0 e^{kt}$
 $V = 36000 e^{kt}$, $V_0 = 36000$
 (i) $\hookrightarrow$ differentiate: $\frac{dV}{dt} = +k V_0 e^{kt} = kV$ ①
 $\therefore$ satisfies,
 (ii) $\frac{1}{4} \times 36000 = 9000$
 $\therefore$ When $t = 30$, $V = 36000 - 9000 = 27000$
 $27000 = 36000 e^{k \times 30}$
 $\frac{27}{36} = e^{30k}$
 $\ln\left(\frac{27}{36}\right) = 30k$ ③
 $k = -0.001547$
 When $t = 60$, $V = 36000 \times e^{-0.001547 \times 60} = 10250$ (17523 if rounded)

(c) (i) $y = \sqrt{9 - x^2}$
 $V = \pi \int_{-3}^3 y^2 dx = 2\pi \int_0^3 (9 - x^2) dx$
 $= 2\pi \left[9x - \frac{x^3}{3} \right]_0^3$
 $= 2\pi [(27 - 9) - 0] = 36\pi \approx 113$ ③
 (ii) sphere ①

(d) $S_n = 3^n + 2n^2$
 $T_{13} = S_{13} - S_{12} = (3^{13} + 2 \times 13^2) - (3^{12} + 2 \times 12^2)$
 $= 1062932$ ②

Question 10

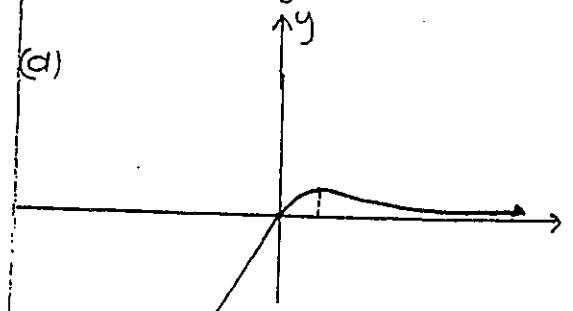
a) $2 \ln x - \ln(2 - x) = \ln 2 - \ln 3$
 $\ln x^2 - \ln(2 - x) = \ln \frac{2}{3}$
 $\ln \frac{x^2}{2 - x} = \ln \frac{2}{3}$
 $\therefore \frac{x^2}{2 - x} = \frac{2}{3}$
 $3x^2 = 4 - 2x$
 $3x^2 + 2x - 4 = 0$
 $x = \frac{-2 \pm \sqrt{4 + 48}}{6}$
 $x = \frac{-2 \pm \sqrt{52}}{6}$
 but $x \neq \frac{-2 - \sqrt{52}}{6}$
 $\therefore x = \frac{-2 + \sqrt{52}}{6} = \frac{-1 + \sqrt{13}}{3}$

b) (i) $e^{-x}(1 - e^{-x}) = e^{-x} - e^{-2x}$
 (ii) (a) $y = e^{-x} - e^{-2x}$
 $= e^{-x}(1 - e^{-x})$
 cut sat $y = 0$
 $e^{-x}(1 - e^{-x}) = 0$
 $e^{-x} \neq 0 \therefore e^{-x} = 1$
 $x = 0$
 $\therefore$ cut sat (0, 0) only

(b) $y' = -e^{-x} + 2e^{-2x}$
 $y'' = e^{-x} - 4e^{-2x}$

s.p. at $y' = 0$
 $-e^{-x} + 2e^{-2x} = 0$
 $e^{-x}(2e^{-x} - 1) = 0$
 $e^{-x} \neq 0, e^{-x} = \frac{1}{2}$
 $-x = \ln \frac{1}{2}$
 $\therefore x = \ln 2$ only soln
 when $x = \ln 2$
 $y = e^{-\ln 2} - e^{-2\ln 2}$
 $= \frac{1}{2} - \frac{1}{4}$
 $y = \frac{1}{4}$
 at $x = \ln 2$
 $y'' = e^{-\ln 2} - 4e^{-2\ln 2}$
 $= \frac{1}{2} - 1 < 0 \therefore$ max
 $(\ln 2, \frac{1}{4})$ is a max

(c) $y' < 0$ when $x > \ln 2$
 as $x \rightarrow \infty$ $y \rightarrow 0$



(d) $A = \int_0^{\ln 2} (e^{-x} - e^{-2x}) dx$
 $= \left[-e^{-x} + \frac{e^{-2x}}{2} \right]_0^{\ln 2}$